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FEDERAL SPECIFICATION

DISHWASHING MACHINES, COMMERCIAL (RACK, STATIONARY)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single tank manually fed, spray type, stationary rack, automatically controlled, commercial dishwashing machines.

1.2 Classification.

1.2.1 Styles, classes, and sizes. Dishwashing machines shall be of the following styles, classes, and sizes, as specified (see 6.2.1).

Style I - Steam heated

Class 1 - Low pressure steam (10 to 15 pound-force per square inch gauge (psig))
Class 2 - High pressure steam (16 to 50 psig), (shipboard use)

Style II - Electrically heated

Size 30-16 - 16- by 16-inch (nominal) racks at 30 racks per hour
Size 40-18 - 18-inch diameter (nominal) racks at 40 racks per hour
Size 45-16 - 16- by 16-inch (nominal) racks at 45 racks per hour
Size 50-20 - 20- by 20-inch (nominal) racks at 50 racks per hour

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

P-D-1801 - Dishwashing Compound, Liquid System (for Commercial Dishwashing Machines)
QQ-N-281 - Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

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QQ-N-288 - Nickel-Copper-Silicon Alloy Castings
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting

Federal Standard:

FED-STD-H28 - Screw Threads for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant, For Treatment of Communications Electronic, and Associated Equipment
MIL-M-11495 - Meter, Detergent Concentration, for Dishwashing Machine
MIL-R-24039 - Rack and Cylinder for Mechanical Dishwashing Machines
MIL-I-43728 - Injector, Rinse, Automatic, for Commercial Dishwashing Machines
MIL-D-43729 - Dispensing Unit, Detergent, Automatic, for Commercial Dishwashing Machines
MIL-K-43875 - Kitchen Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Federal Regulations:

Occupational Safety and Health Administration (OSHA):

Occupational Safety and Health Standards

"(The Code of Federal Regulations and the Federal Register are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)"

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials (ASTM):

A120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless for Ordinary Use
A167 - Stainless and Heat Resisting Chromium Nickel Steel Plate, Sheet and Strip
A436 - Austenitic Gray Iron Castings
B302 - Threadless Copper Pipe

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute, Inc. (ANSI):

S1.4 - Specification for Sound Level Meters
S1.13 - Methods for the Measurement of Sound Pressure Levels

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Electrical Manufacturers Association (NEMA):

ICS 2 - Industrial Controls and Systems
MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

National Fire Protection Association (NFPA):

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

National Sanitation Foundation (NSF):

No. 3 - Dishwashing Machines/Commercial Spray Type
No. 5 - Hot Water Generating Equipment

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(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, 3475 Plymouth Road, Ann Arbor, MI 48106.)

Society of Automotive Engineers, Inc. (SAE):

J534 - Lubrication Fittings

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

Underwriters Laboratories Inc. (UL):

UL 921 - Commercial Electric Dishwashers
UL Electrical Appliance and Utilization Equipment List

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The dishwashing machines shall consist of the following principal parts: Base (frame), legs, wash chamber hood, tank, spray assemblies, pumps, motors, controls, piping, valves, heating equipment and accessories.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a complete dishwashing machine for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Safety and health requirement. The machine shall be equipped with safety devices for all parts that present safety hazards. The devices shall include cover and guards for moving parts and shockproof controls for protection from mechanical and electrical hazards to personnel. All guards shall provide easy access to guarded parts and shall not interfere with operation of the machine, and conform to OSHA, Title 29, Chapter XVII, Part 1910 (see 4.5.3, 6.2.1, and 6.5).

3.4 Standards compliance. Dishwashing machines shall conform to UL 921 and NSF No. 3 and No. 5.

3.4.1 Certification. Prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dishwashing machines he proposes to supply under this specification meets the requirements of UL 921, NSF No. 3 and No. 5.

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3.4.1.1 UL certification. Acceptable evidence of meeting the requirements of UL 921 shall be the UL label, listing in the UL Electrical Appliance and Utilization Equipment List, or a certified test report from a nationally recognized independent testing laboratory, acceptable to the contracting officer, stating that the dishwashing machines offered have been tested and conform to UL 921.

3.4.1.2 NSF certification. Acceptable evidence of meeting the requirements of NSF No. 3 and No. 5 shall be the NSF Seal on the finished dishwashing machine, listing in the NSF Seal of Approval Listing of Food Service Equipment (see 3.8), or a certified test report from a recognized independent testing laboratory, acceptable to the medical department of the procuring activity, indicating that the dishwashing machines conform to NSF No. 3 and No. 5.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6.1 Corrosion-resisting steel. Corrosion-resisting steel shall conform to the chemical composition requirements of any 300 series stainless steel specified in QQ-S-766.

3.6.2 Nickel-copper alloy. Nickel-copper alloys shall conform to the requirements of QQ-N-281 or QQ-N-288.

3.6.3 Galvanized steel. Galvanized steel shall conform to ASTM A120.

3.7 Design. The dishwashing machines shall be so designed as to facilitate cleaning of all parts, permit complete drainage of water, uniformly wash all dishes, and eliminate entirely, or reduce to a minimum, the areas where dirt and dust might accumulate and provide shelter for bacteria and vermin. The pump motor shall not extend beyond the dimensions of the hood. The machines shall be designed either for straight-through operation or for corner operation as specified (see 6.2.1). Dishwashers shall have an inside working height, including door height, of not less than 10 inches above the top of the track except that the size 50-20 machine shall have a height of not less than 16 inches.

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3.7.1 Water temperature, pressure, and volume. The dishwashing machines shall automatically maintain the water temperature and provide water of adequate volume, pressure, and cutting velocity, as specified in NSF No. 3 and No. 5. Unless otherwise specified (see 6.2.1), the dishwashing machines shall be designed for satisfactory operation with a supply of water and steam, as applicable under the following conditions: Water supply pressure - 20 to 80 psig, hot water temperature 60 degree Celsius (°C) (140 Fahrenheit (°F)), steam supply pressure - class I - 10-15 psig, class II - 16-50 psig.

3.8 Performance. When operated as prescribed by the manufacturer, the dishwashing machine shall be capable of rendering dishes (multiuse eating and drinking utensils) physically and hygienically clean, and in a sanitary condition to prevent transmission of disease. Compliance with this requirement shall be evidenced by sanitation certification (see 3.4.1.2). The operating capacity for the dishwashing machine shall be not less than that indicated in paragraph 1.2.1 for the applicable style and size.

3.8.1 Noise level. Unless otherwise specified (see 6.2.1 and 6.5), dishwasher noise level shall not exceed 80 A-weighed decibels at the two operator positions. The operator's positions are 1-1/2 feet from the loading end of the dishwasher, 1-1/2 feet from the unloading end and 5 feet above the floor (see 4.5.3).

3.9 Construction. The dishwashing machine shall be complete so that when connected to the specified source of power, water supply, heating means (steam or electric), and drainage, as applicable, the unit can be used for its intended function. Dishwashers shall be rigid, quiet in operation, free from vibration, and so constructed as to prevent splashing of water or overflow of water to the outside of the machine. Parts requiring adjustment or lubrication shall be readily accessible from the front of the machine. The machine shall wash dishes by means of a water and detergent solution pumped from a tank, and shall final rinse the dishes with fresh water from an outside source. Provisions shall be made to fill the wash tank either directly from the regular hot water supply with a hand valve or through the booster and solenoid. The wash, dwell, and rinse cycles shall be automatic and electrically controlled. A light shall be provided to indicate when the machine is in operation. The rinse spray control shall have a positive return to the off position. Machines shall be provided with tracks of corrosion-resisting steel not less than No. 16 gauge (0.060 inch).

3.9.1 Fastening devices. All screws, pins, rivets, nuts, washers, and similar parts shall be installed with means for preventing loss of tightness. When subject to removal or adjustment such parts shall not be swagged, peened, staked, or otherwise permanently deformed. Steel fasteners shall be used except when dishwashing machines have been fabricated of metals other than steel, in which case the fasteners shall be of the same material as the fastened members.

3.9.2 Piping, copper tubing, fittings, and valves. All piping, copper tubing (copper tubing shall conform to ASTM B302), fittings, and valves for the control of water, drainage, overflow, and steam shall be installed by the supplier and shall be so arranged that a minimum of pipeline connections will be necessary. Connections shall be readily accessible to facilitate

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installation. Piping, tubing, and valves shall be located, whenever possible, on the exterior of the machine. Gate valves used shall be located with stems above horizontal plane of the piping.

3.9.3 Piping and fittings. Water, steam piping, and fittings shall be of stainless steel, copper, or brass. Fresh water supply to the tank shall be discharged not lower than 2 inches above the maximum flood level rim, or an effective vacuum breaker may be installed to prevent backflow. A vacuum breaker shall be installed on the final rinse line. The drain and other connections shall be standard weight (schedule 40) pipe or tubing connections. Drainage piping shall be stainless steel, galvanized steel, or copper, with drainage fittings. Drains may be joined into a single trun' line requiring only one connection or arranged to permit individual connections to the waste line. Pump intake and discharge systems shall be of stainless steel, nickel-copper alloy, bronze castings, or black iron.

3.9.4 Valves. Steam valves shall be bronze, full-plug type, with seat and disk of stainless steel with a minimum Brinell hardness of 500, and designed for a saturated steam working pressure of 125 psig. A valve shall be provided for filling the tank. The drain valve shall be a shutoff type, and permanently marked to show "open" and "closed" position and shall be lever-operated or wheel-operated, ruggedly designed for foot or hand operation. A line strainer and a pressure-reducing valve, factory set at 20 psig discharge pressure flowing, shall be furnished and properly installed in the hot water supply line to the final rinse. Fresh water rinse valves shall be fully automatic, packless, piston-pilot-operated, or diaphragm solenoid valves, normally in the closed position and suitable for 100°C (212°F) water. Numbered metal, plastic or cardboard tags shall be attached to the valves for easy identification.

3.9.5 Pumped spray assembly. All spray nozzles and spray arm manifolds shall be corrosion-resisting metal slotted or fitted with corrosion-resisting metal or high impact plastic nozzles. All spray assemblies shall be removable without the use of tools and shall be easy to clean. Spray assemblies shall be fitted with easily removable end caps or other closures that permit ready insertion of cleaning brushes. Except where revolving type spray assemblies are furnished, top wash and rinse manifolds shall each have a minimum of three fixed spray arms, and bottom wash and rinse manifolds shall each have a minimum of four fixed spray arms.

3.9.6 Tank. The tank shall be constructed of not less than 0.0598-inch (US revised standard gauge No. 16), stainless steel, conforming to ASTM A167, and shall be watertight, and of welded or deep drawn construction. Visible water level indicators shall be installed to show the level of water in the tank.

3.9.6.1 Overflow. The dishwashers shall have an accessible overflow drain in the tank. In the overflow unit at least 6 inches of skimming surface for 16-inch rack machines and 10 inches of skimming surface for 20-inch rack machines shall effectively free the water from scum. The overflow unit shall be readily removable or have a protective cover attached to the overflow pipe. Overflow units shall be at least 1 inch below the bottom of scrap trays.

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3.9.7 Scrap trays (strainer). Scrap trays of stainless steel not less than 0.0359-inch (US revised standard gauge No. 20) thick, shall be provided to prevent insoluble matter and large pieces of food residue from passing into the tank. No opening around or between the scrap trays shall be wider than the diameter of the openings in the scrap trays. The ledges on which the scrap trays rest shall be so designed that surfaces beneath the ledges are easily accessible for cleaning when the trays are removed.

3.9.8 Vents and doors. Vents, doors, and door frames shall be constructed of not less than 0.0478-inch (US revised standard gauge No. 18) stainless steel; shall be rigid and stiffened as necessary. The hoods of the roll door design shall be 0.0359-inch (US revised standard gauge No. 20) minimum. When specified (see 6.2.1), exhaust vent collars with internal baffles shall be installed directly to the hood. When of the vertical opening type, loading and unloading doors shall be counterbalanced and when in the raised position shall electrically interlock the machine so that it cannot be started accidentally. Opening the door during operation shall automatically stop the machine. Automatic door catches shall be provided on doors not counterbalanced. Chain operated counterbalanced door, when provided, shall have the door chain fastened at the counterweight by means of an idler pulley. Doors shall be splashproof and their exposed edges shall be smooth and formed to prevent canting or warping.

3.9.9 Legs (feet). The machine shall have NSF approved construction, with four or more legs (feet) made of corrosion-resisting steel having polished surfaces or corrosion-resisting material. Legs shall be adjustable, so that the height of the track may be varied from 34 to 35 inches above the floor.

3.9.10 Pump and motor assemblies.

3.9.10.1 Assemblies. The pump and motor shall be mounted on a rigid steel base under the track or on the tank. Rotary seals shall be provided for pump shafts and shall be readily accessible for servicing.

3.9.10.2 Pump. Pump casings shall be cast iron or stainless steel and shall have a removable cover or inspection plate, or be of such a design as to permit easy accessibility for inspection and cleaning of impeller and interior. The pump shall either be self-draining or equipped with means for draining; the arrangement shall be such that when the tank is drained the pump will also be drained. The shaft shall be of stainless steel, properly alined and supported. The impeller shall be stainless steel or gray iron alloy conforming to ASTM A436 and shall be balanced statically and dynamically. The pump shall have at least two ball or roller bearings, except that when the pump and motor are mounted on the same shaft, at least two ball or roller bearings shall be provided for the motor and pump. The pump suction intake shall be provided with a stainless steel strainer.

3.9.11 Thermometers. Dishwashing machines shall be fitted with replaceable, dial-type, watertight thermometers readily visible from the front of the machine and positioned to prevent breakage. The thermometers shall have diameters of 3 inches or larger. Two thermometers shall be provided to indicate the wash and rinse water temperatures. The words "WASH" and "FINAL RINSE" shall be etched, stamped, or engraved on corrosion-resisting steel plates and affixed to the machine in close proximity to the thermometers, or

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the words may be printed on the thermometers. Thermometers shall be graduated to at least 100°C (212°F) and shall be accurate within ±2°C (±3°F). Thermometers shall be marked to indicate the proper temperature range.

3.9.12 Pressure gauges. One pressure gauge shall be mounted on the discharge side of the pressure reducing valve to indicate final rinse pressure and one shall be mounted in a convenient location to indicate the wash pressure at the top of the spraying manifold. Pumped wash gauge dials shall be colored red below markings indicating 4 pound-force per square inch (psi) and above 6 psi. All pressure gauges shall be easily replaceable, dial-type having diameters 2 inches or larger and be equipped with pulsation dampeners.

3.9.13 Heating. Style I, class 1, dishwashing machines shall be suitable for operation with a steam supply working pressure of from 10 to 15 psig. Style I, class 2, dishwashing machines shall be suitable for operation with a steam supply working pressure of from 16 to 50 psig. The style II dishwashing machine shall be capable of satisfactory operation, utilizing the electric power supply specified herein.

3.9.13.1 Heat exchanger. Style I dishwashers shall be provided with an adjustable, automatic, heat exchanger to raise the temperature of the final rinse water from 55°C (130°F) to at least 86°C (180°F). The heat exchanger shall automatically maintain the required final rinse water temperature without producing steam within either the heat exchanger or the water supply piping from the heat exchanger to the machine. The heat exchanger shall be securely mounted as an integral part of the machine in a position that does not interfere with operation and permits attachment of tables or counters. Otherwise, the heat exchanger shall be furnished separately, mounted on its own legs and equipped with suitable fittings for connection into the final rinse water lines. The heat exchanger shall be provided with means for complete drainage of the steam jacket and water tubes. Required valve and the temperature regulator shall be accessible and adjustable from the front of the machine. Valves and pipe unions shall be installed on the heat exchanger where the steam and water lines enter the unit. The final rinse water temperature shall be controlled by an automatic thermostat. The medium transmitting from the heat sensing bulb to the regulator shall be protected by stainless steel, armored, flexible tubing. A steam thermostatic and float trap shall be installed on the condensate line.

3.9.13.2 Electric booster, style II. The style II dishwashers shall be provided with an electric booster having all necessary controls for automatic operation to raise the temperature of the final rinse water from 55°C (130°F) to at least 86°C (180°F). The electric booster shall be designed to operate with the electric power characteristics specified (see 6.2.1). The electric booster shall be UL approved.

3.9.13.3 Water tank heating. Steam heat exchangers with temperature thermostats shall be provided for maintaining the proper water temperature in the pumped wash tank. When electric heating is specified (see 3.9.13), electric heaters with thermostats and low-water cutoffs shall be installed. Heater elements and sheaths shall be 300 series stainless steel.

3.9.14 Detergent control. Unless otherwise specified (see 6.2.1), the dishwashing machine shall be furnished with an electric or electronic automatic detergent dispenser conforming to MIL-D-43729. The reservoir shall

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be mounted in a convenient, accessible position and shall be capable of holding a supply of powdered dishwashing compound conforming to type I of P-D-1801 or liquid detergent at least equal to the amount used in a normal dishwashing operation for one meal (2-11 pounds of powdered compound).

3.9.15 Rinse agent injector. When specified (see 6.2.1), a rinse agent injector conforming to MIL-I-43728 shall be supplied with each dishwashing machine and shall be installed without causing interference with operating maintenance or cleaning procedures.

3.9.16 Detergent concentration meter. When specified (see 6.2.1), a detergent concentration meter conforming to MIL-M-11495 and all accessories necessary for proper operation shall be supplied with each machine but not installed. The machine shall be drilled for mounting the electrodes in the wash tank and the meter on top of the machine, as illustrated in MIL-M-11495. This requirement is intended to apply to machines procured for remote installation sites where service support for automatic detergent dispensers would not be available and manual addition of washing compounds is anticipated.

3.10 Electrical equipment. Unless otherwise specified herein, electrical equipment shall meet the requirements of UL 921. The dishwasher shall operate on the power characteristics (current, voltage, phase, frequency) specified (see 6.2.1).

3.10.1 Motors. Unless otherwise specified (see 6.2.1), motors shall be dripproof encapsulated, or sealed windings. Motors shall be designed and rated for continuous duty in an ambient temperature of 50°C (122°F). The motors shall be polyphase, squirrel-cage inductor type, or single phase induction type conforming to NEMA MG 1. The continuous horsepower ratings of the motors shall meet the pumping rates at the head pressure in the spray assemblies to meet the washing and rinsing action required under NSF No.3 and No. 5.

3.10.2 Controls. All control equipment shall conform to NEMA ICS 2 and be designed for operation in an ambient temperature of 50°C ±5°C (122°F ±9°F). Thermal overload protection of the manual or automatic reset type shall be provided. Switches and controls exclusive of thermostat and solenoid valves shall be mounted on the machine in NEMA type 12 enclosures, and shall be conveniently located for use by an operator. Controls and switches shall be so arranged that all line leads shall be open in the off position. When specified (see 6.2.1), suitable controls shall be installed to prevent operation of the machine unless the final rinse water temperature is 86°C (180°F) or higher.

3.10.3 Wiring and circuit safety devices. All wiring and circuit safety devices shall be in conformance with NFPA No. 70. All wiring between the dish-washing machine components shall be complete to a common junction box, requiring a minimum number of connections to the main electrical power supply.

3.11 Equipment.

3.11.1 Standard equipment. Unless otherwise specified (see 6.2.1), each dishwashing machine shall be furnished with the applicable number of racks shown in table I and with three brushes of suitable size and type for use in

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cleaning spray assemblies. Racks and cylinders shall conform to the requirements of MIL-R-24039.

3.11.2 Accessory equipment, spare and service parts. Accessory equipment, spare and maintenance parts shall be furnished as specified (see 6.2.1).

TABLE I. Dishwashing machine racks and cylinders.

Nominal size inches	Types															Cylinders	
	I	IA	IB	IC	ID	IE	IF	IG	II	IIA	IIC	IID	IIE	III	IIIB	V	VI
30-16	-	-	-	-	2	2	-	-	-	-	-	2	-	-	-	8	8
45-16	-	-	-	-	2	2	-	-	-	-	-	2	-	-	-	8	8
40-18	-	-	-	-	-	-	2	-	-	-	-	-	2	-	-	-	-
50-20	4	-	-	-	-	-	-	-	3	-	-	-	-	2	-	12	12
50-20	4*	6	-	2	-	-	-	4	6	8	4**	-	-	-	2	12	12

(For Army only)

* For confinement and training dining facilities

**Style B rack (24 or 25 compartments)

3.12 Lubrication. Unless otherwise specified (see 6.2.1), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.14 Identification marking. Identification shall be permanently and legibly marked directly on the dishwashing machine or on a corrosion-resisting metal plate securely attached to the machine at the source of manufacture.

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Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer. In addition, information required by UL 921 shall be included on the dishwasher or on the plate.

3.15 Electromagnetic compatibility. When specified (see 6.2.1), equipment procured under this specification shall be designed and equipped to meet the electromagnetic interference control requirements and test limits for class C3, group III equipment as specified in MIL-STD-461 and MIL-STD-462.

3.16 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.17 Commercial publications. Manufacturer's standard commercial publications shall be furnished (see 6.2.1).

3.18 Workmanship.

3.18.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.18.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.18.3 Solder. All surfaces to be soldered (when applicable) shall be thoroughly cleaned prior to soldering. Solder joints shall be smooth with no pin holes.

3.18.4 Threaded fasteners. Threaded fasteners shall not be loose or missing. Threads shall not be missing, stripped, or broken, and shall conform to FED-STD-H28.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or

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any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.1.2 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standards cited in 3.4.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete dishwashing machine when a first article is required (see 3.2, 6.2.1, and 6.4). This inspection shall include the examination of 4.4, and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examinations of 4.4, the tests of 4.5.1 through 4.5.3, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All dishwashing machines offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level II and an AQL of 4.0 percent defective.

4.4 Examinations.

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4.4.1 Visual examination. Each sample selected shall be examined for defects listed in table II. Where multiple defects are grouped together (e.g., electrical assembly) each deficiency (e.g., were not properly joined, not enclosed in conduit) shall constitute one defect.

4.5 Tests.

4.5.1 Operational test. Each sample machine shall be operated at least 1 hour without dishes or racks to determine compliance with the requirements of this specification. The test shall reveal that the assembly of component parts is complete and properly interrelated; that moving parts evidence no undue vibration, noise, or overheating of bearings, and that controls operate smoothly and positively.

4.5.2 Leak test. The sample dishwashing machines shall be filled with water as required and operated in accordance with manufacturer's instructions for a period of at least 10 minutes. The tanks, pipes, valves, traps, strainers, and fittings shall show no evidence of leakage. This test may be performed concurrently with the test of 4.5.1.

4.5.3 Noise level test. Compliance with the noise level limits of 3.8.1 shall be verified by measurement. Noise levels shall be measured using a sound level meter in accordance with ANSI S1.4, type 1. The meter shall be set for the A-weighting "slow" mode of operation. The noise level test shall be set up and conducted in accordance with the guidelines specified in ANSI S1.13 for the "Field" measurement method. The dishwashing machine shall be operated with a maximum number of dishes and racks as per manufacturer's recommended maximum amount. The noise level measurement shall be made at the loading end and at the unloading end of the dishwasher at the operator's position. Failure to meet the noise level requirements of 3.8.1 shall be cause of rejection. This test may be performed concurrently with the test of 4.5.1 and requirements of 3.3.

4.5.4 Electromagnetic interference control tests. When electromagnetic interference control is required, the first article sample, equipped for the reduction of electromagnetic interference in accordance with 3.15, shall be tested by the supplier in accordance with test methods UM-05 of MIL-STD-462. The supplier shall furnish the contracting officer, for approval, the interference control plan, the EMI-EMC test plan and the test report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the first article sample shall be used as a model for all other units.

4.6 Preparation for delivery inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-K-43875.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The preservation, packaging, and packing shall be in accordance with MIL-K-43875. The levels of preservation and packaging, and level of packing shall be as specified (see 6.2.1).

5.2 Marking.

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5.2.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The dishwashing machines covered by this specification are intended to be used to uniformly wash and rinse eating and drinking utensils in installations requiring large-scale service of food to Governmental employees (Federal and Military).

6.2 Ordering data.

6.2.1 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification
b. Style, class, and size machine required (see 1.2.1)
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4)
d. When safety and health requirements are different, specify and fully describe (see 3.3 and 6.5)
e. Whether machine shall be designed for straight-through or corner operation (see 3.7)
f. Water and steam supplies, if other than specified (see 3.7.1)
g. Noise level requirements, if other than specified (see 3.8.1)
h. When exhaust vents are required (see 3.9.8)
i. Electrical power supply characteristics (current, voltage, phase, frequency) (see 3.9.13.2 and 3.10)
j. When a detergent dispenser is required (see 3.9.14)
k. When a rinse agent injector is required (see 3.9.15)
l. When a detergent concentration meter is required (see 3.9.16)
m. Motors, if other than specified (see 3.10.1)
n. When controls are required to prevent operation when final rinse temperature drops below 180°F (see 3.10.2)
o. Standard equipment, if other than specified (see 3.11.1)
p. Accessory equipment, spare and maintenance parts required (see 3.11.2)
q. When other means of lubrication are required (see 3.12)
r. Treatment and painting if other than specified (see 3.13)
s. When electromagnetic interference control is required (see 3.15)
t. When fungus-resistance coating of electrical circuit elements is required (see 3.16)
u. When commercial publications are required (see 3.17)
v. Level of preservation and packaging and level of packing required (see 5.1)

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an

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approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required (see 3.2), the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one dishwashing machine. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the OSHA regarding emission, noxious vapors, and heat and noise level requirements, as applicable, specific requirements concerning guarding the points of operation, specific requirements concerning guarding the point(s) of operation and other safety and health requirements should be specified (see 3.3, 4.5.3, and 6.2.1).

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - YD
Air Force - 99

Review Activities

Army - MD
Navy - SA
Air Force - 84

User Activities

Navy - MS, CG, MC
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - OPP
HHS - FDA
EPA - PM-215

PREPARING ACTIVITY:

Navy - YD

DoD Project 7320-0731

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TABLE II. Classification of Defects.

Classification	Defects	Requirement paragraph
Major:		
101	Dishwashing machines not of specified classification	1.2 and 3.1
102	Units in a lot not identical to the extent required to insure interchangeability of support parts	3.5
103	Material not of type specified	3.6
104	Defects in materials affecting usability of the unit	3.6
105	Design not as specified	3.7
106	Dimensions not within specified limit	3.7
107	Operation omitted or not properly performed	3.8
108	Part missing or not specified type or size	3.9
109	Part fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired	3.9
110	Adjustable assembly that cannot be adjusted to perform the function intended	3.9
111	Component not properly assembled or secured affecting serviceability; e.g., tubing creased at bend restricting or partially restricting interior flow	3.9
112	Not in conformance with applicable requirements	3.9
113	Part misplaced, loose, or not in proper alinement affecting serviceability	3.9.1
114	Service parts not furnished when specified	3.11.2
115	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	3.11.2
116	Lubrication means for not provided or inaccessible	3.12
117	Fungus resistance not applied when specified	3.16
Minor:		
201	Functioning component requiring abnormal force to operate	3.9
202	Components not connected or joined as specified or poorly accomplished	3.9
203	Component not readily accessible for servicing where required	3.9.2
204	Marking missing or incomplete	3.14
205	Marking missing or illegible	3.14
206	Finish not as specified	3.13
207	Finish application incomplete	3.13
208	Instruction plate Missing, Incomplete or illegible	3.14
209	Publications not furnished or incomplete	3.17

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